

Field Effect Transistors (FET)

▣ Definition of JFET.

Ans: A Junction Field Effect Transistor (JFET) is defined as a voltage controlled semiconductor device that controls current flow using majority carriers.

▣ Draw the symbol of N-channel JFET and P-channel JFET.

Ans:

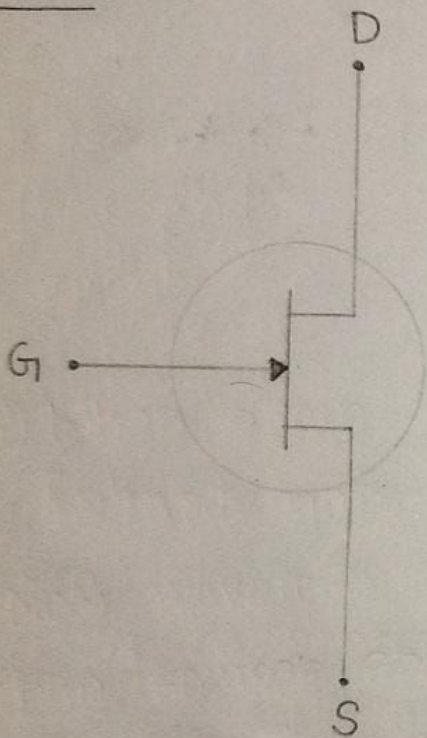


Fig: n-channel JFET

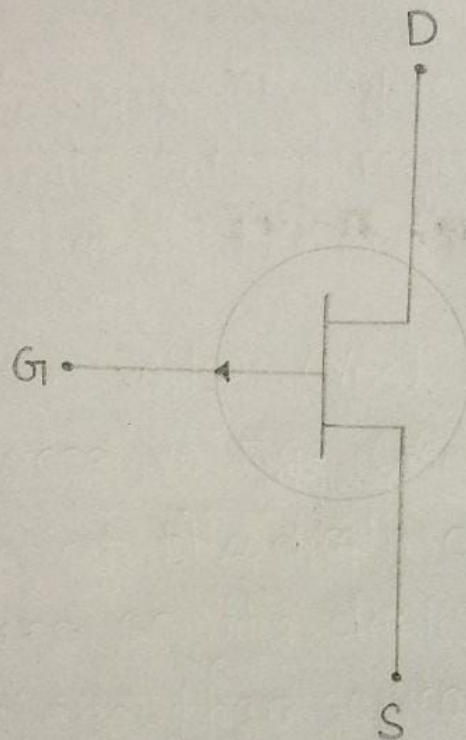


Fig: p-channel JFET

Q Describe construction and working principle of (N-channel JFET).

Ans: Construction of JFET:

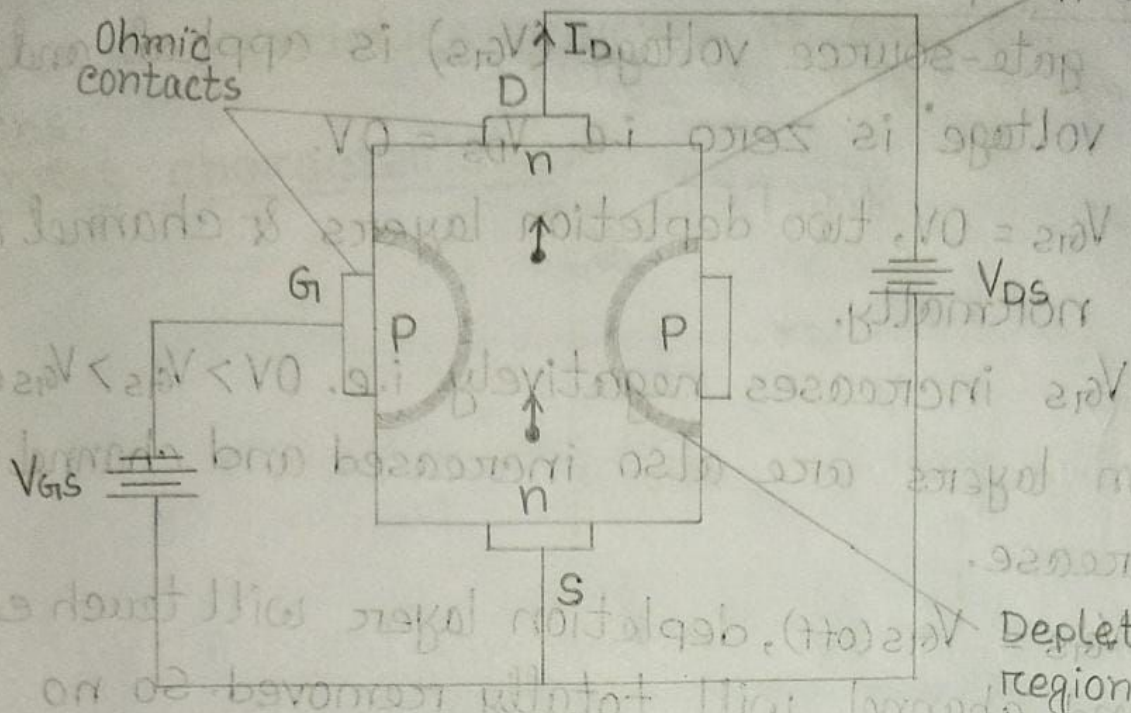


Fig: n-JFET

Source: The terminal through which the majority carriers enter into the channel, is called the source terminal S.

Drain: The terminal through which the majority carriers leave from the channel, is called the drain terminal D.

Gate: There are two internally connected heavily doped impurity regions to create two P-N junctions. These impurity regions are called the gate terminal G.

a) Channel: The region between the source and drain, sandwiched between the two gates is called the channel.

Theory of operation:

(i) When gate-source voltage (V_{GS}) is applied and drain source voltage is zero i.e. $V_{DS} = 0V$.

- When $V_{GS} = 0V$, two depletion layers & channel are formed normally.
- When V_{GS} increases negatively i.e. $0V > V_{GS} > V_{GS(off)}$, depletion layers are also increased and channel will be decrease.
- When $V_{GS} = V_{GS(off)}$, depletion layer will touch each other and channel will totally removed. So no current can flow through the channel.

(ii) When drain-source voltage (V_{DS}) is applied at constant gate-source voltage (V_{GS}).

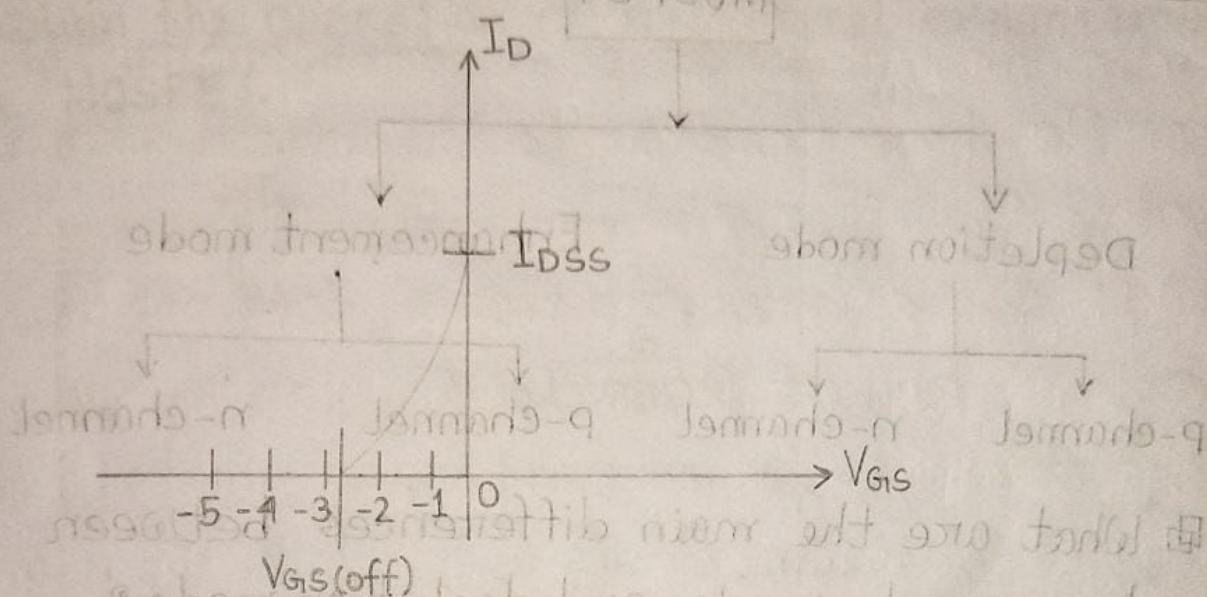
- Now reverse bias at the drain end is larger than source end and so the depletion layer is wider at the drain end than source end.
- When V_{DS} increases i.e. $0V < V_{DS} < V_p$, depletion layer at drain end is gradually increased and drain current also increased.

- When $V_{DS} = V_p$ the channel is effectively closed at drain end and it doesn't allow further increase of drain current. So, the drain current will become constant.

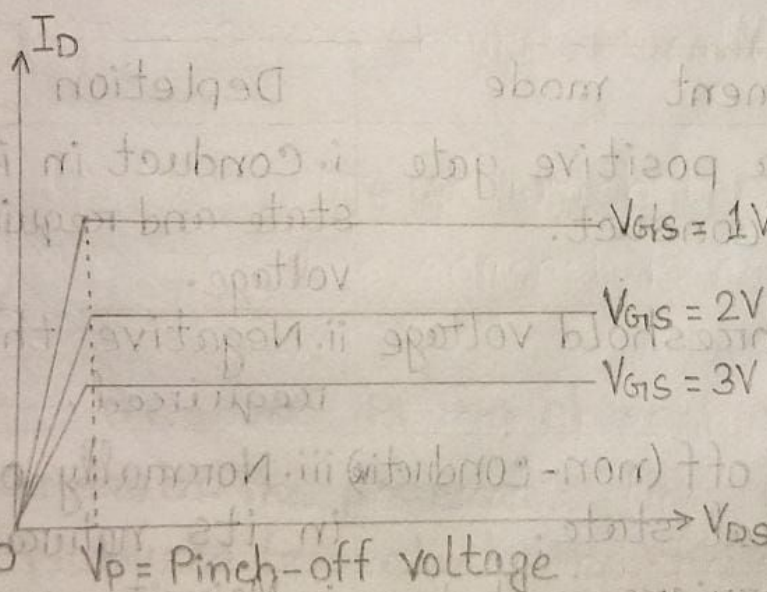
▣ characteristics curves of n-JFET.

Ans:

input characteristics



output characteristics

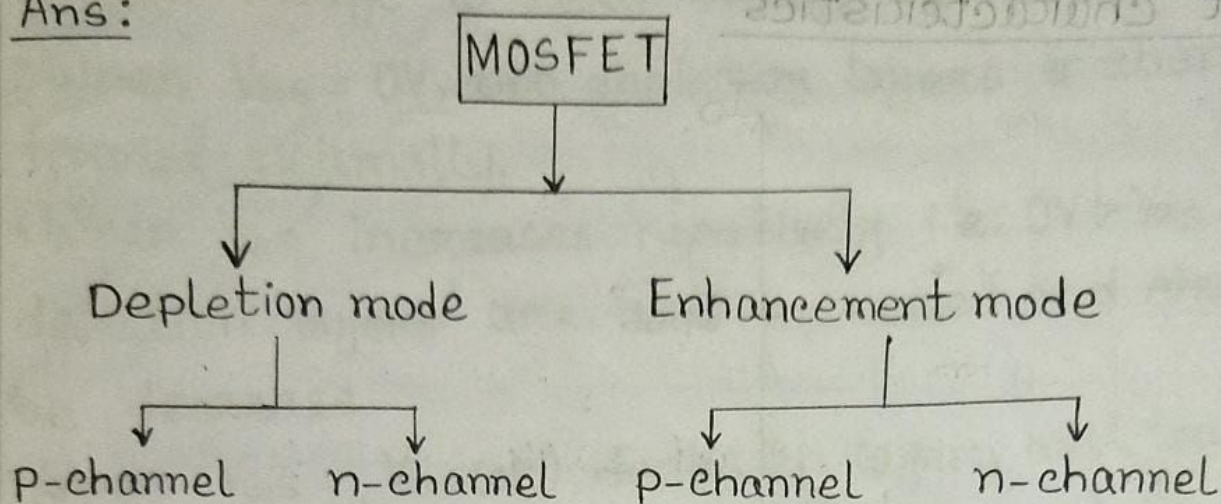


Q What is pinch-off voltage in JFET's?

Ans: It is the minimum drain-source voltage at which the drain current essentially becomes constant.

Q Different types of MOSFET.

Ans:



Q What are the main differences between enhancement-mode and depletion-mode?

Ans:

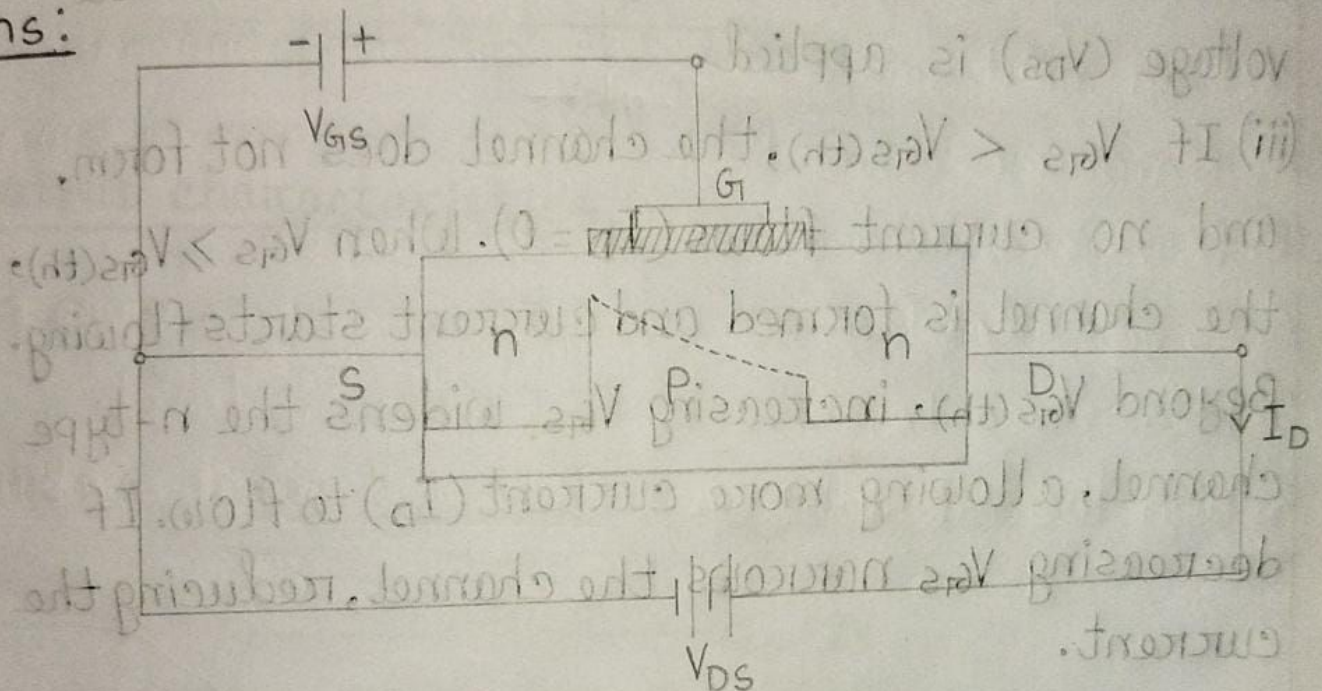
Enhancement mode	Depletion mode
i. Requires a positive gate voltage to conduct.	i. Conduct in its natural state and requires no gate voltage.
ii. Positive threshold voltage required.	ii. Negative threshold voltage required.
iii. Normally off (non-conductive) in its natural state.	iii. Normally on (conductive) in its natural state.
iv. Majority carriers are electrons.	iv. Majority carriers are holes.

▣ Definition of MOSFET.

Ans: The MOSFET (Metal Oxide Semiconductor Field Effect Transistor) transistor is a semiconductor device which is widely used for switching and amplifying electronic signals in the electronic devices.

▣ Explain the operation of n-channel enhancement mode MOSFET.

Ans:



Theory of operation:

(i) When $V_{GS} = 0V$, there is no channel connecting the source and drain. The p substrate has only a few thermally produced free electrons (minority carriers) so that drain current is essentially zero. For this reason,

E-MOSFET is normally OFF.

(ii) When $V_{GS} > 0V$, a positive voltage is applied to the gate. This positive voltage attracts free electrons in the p-substrate toward the gate (just beneath the SiO_2 layer), forming a n-type channel between the source and the drain. When this channel is formed, current (I_D) can flow from the drain to the source when a voltage (V_{DS}) is applied.

(iii) If $V_{GS} < V_{GS(th)}$, the channel does not form, and no current flows ($I_D = 0$). When $V_{GS} \geq V_{GS(th)}$, the channel is formed and current starts flowing. Beyond $V_{GS(th)}$, increasing V_{GS} widens the n-type channel, allowing more current (I_D) to flow. If decreasing V_{GS} narrows the channel, reducing the current.

▣ What is threshold voltage in MOSFET?

Ans: The minimum value of V_{GS} that turns the E-MOSFET ON is called threshold voltage [$V_{GS(th)}$].

▣ Characteristics curves of n-EMOSFET.

Ans: input characteristics

I_D

V_{GS}

V_{TH} = Threshold Voltage

output characteristics

I_D

$V_{GS} = 3V$

$V_{GS} = 2V$

$V_{GS} = 1V$

V_{DS}

0

Q Write down the differences between JFET and BJT.

Ans:

JFET	BJT
1. JFET is a voltage-controlled device.	1. BJT is a current-controlled device.
2. JFET has a single doped semiconductor material without any pn junction.	2. BJT has three doped semiconductor layers, forming two pn junctions.
3. High input impedance.	3. Low input impedance.
4. Unipolar (majority).	4. Bipolar (majority & minority).
5. Low-medium gain.	5. High gain.
6. It consumes less power.	6. It consumes more power.
7. Low noise level.	7. High noise level.
8. High thermal stability.	8. Low thermal stability.

Example: Fig.1 shows the transfer characteristics curve of a JFET. Write the equation for drain current.

Solution: Referring to the transfer characteristics curve in fig.1, we have,

$$I_{DSS} = 12\text{mA}, V_{GS(\text{off})} = -5\text{V}$$

$$\therefore I_D = I_{DSS} \left[1 - \frac{V_{GS}}{V_{GS(\text{off})}} \right]^2$$
$$= 12 \left[1 + \frac{V_{GS}}{5} \right]^2 \text{mA.}$$

Example: A JFET has the following parameters: $I_{DSS} = 32\text{mA}$; $V_{GS(\text{off})} = -8\text{V}$; $V_{GS} = -4.5\text{V}$. Find the value of drain current.

Solution:

$$I_D = I_{DSS} \left[1 - \frac{V_{GS}}{V_{GS(\text{off})}} \right]^2$$
$$= 32 \left[1 - \frac{(-4.5)}{-8} \right]^2 \text{mA}$$
$$= 6.125 \text{mA.}$$

Example: A JFET has a drain current of 5mA. If $I_{DSS} = 10 \text{ mA}$ and $V_{GS(off)} = -6 \text{ V}$, find the value of (i) V_{GS} and (ii) V_p .

Solution: $I_D = I_{DSS} \left[1 - \frac{V_{GS}}{V_{GS(off)}} \right]^2$

or, $5 = 10 \left[1 + \frac{V_{GS}}{6} \right]^2$

or, $\left[1 + \frac{V_{GS}}{6} \right]^2 = 0.5$

or, $1 + \frac{V_{GS}}{6} = 0.707$

or, $\frac{V_{GS}}{6} = -0.293$

(i) $\therefore V_{GS} = -1.76 \text{ V}$

(ii) $V_p = -V_{GS(off)} = -(-6) = 6 \text{ V}$